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What's special about 89% credibility intervals?

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A curious information-theoretic property of 89% credibility intervals is discussed.

A *credibility interval* (or "credible", "Bayesian", "posterior", "compatibility" interval or region) is a range where the true value of some real quantity lies with a given probability[1]. For instance, if $[1.2, 7.0]$ is a 60% credibility interval for the quantity Q, then there is a 60% probability that the true value of Q is between 1.2 and 7.0 (note the difference in meaning from a *confidence* interval, whose definition is much more involved and less direct[2]). It is always best to report the full posterior probability distribution of a quantity of interest, so that the reader can calculate from it the probabilities of any desired ranges and make decisions[3]. But credibility intervals often provide useful summaries of the full distribution.

Credibility intervals of 50%, 75%, 90%, 95%, or even 99% probability are more or less commonly used in the Bayesian literature and given as defaults in software packages for Bayesian analysis. Also *89%* ones have become quite fashionable since around 2016; they seemingly originate in McElreath's *Statistical Rethinking*[4], first published in that year. They are used, for example, in the software package *bayestestR*[5]. McElreath justly points out[6] that the specific probability value of a credibility interval should be chosen depending on the context; there is nothing special about values like 90% or 95%. To emphasize and remind us of this fact (and also to avoid confusion with confidence intervals, whose meaning is different) he uses 89%, which is an arbitrary value.

Or is it?

It turns out that 89% credibility intervals do possess a curious information-theoretic property, which may also motivate the use of 50% and 89% as defaults.

[1] Bernardo & Smith 2000 § 5.1.5 pp. 259ff. [2] MacKay 2005 § 37.3; Jaynes 1976. [3] Raiffa 1970. [4] McElreath 2020. [5] Makowski, Ben-Shachar, et al. 2019. [6] McElreath 2020 e.g. § 3.2.2 p. 56, § 4.3.5 p. 88.

The true value of a quantity lies inside a p credibility interval with probability p, and outside with probability $1 - p$. Our uncertainty about these two possibilities can be quantified by the Shannon entropy:[7]

$$H(p) \coloneqq -[p \log_2 p + (1-p) \log_2 (1-p)] \,\mathrm{Sh}\,. \tag{1}$$

The value of 1 Sh (one shannon[8]) represents an equal uncertainty between two alternatives.

Thus for a 50% credibility interval we have an uncertainty of 1 Sh on whether the true value lies inside or outside it. This is the highest level of uncertainty for such dichotomous alternative. It is useful to also provide an interval with an uncertainty smaller than 1 Sh, so that we are somewhat more certain that the true value lies inside than outside.

For example, for which interval is our maximal uncertainty *halved* to 0.5 Sh? The answer is: for *an* 89.00% *credibility interval*. This result is easily calculated solving $H(p) = 0.5\,\mathrm{Sh}$ numerically. A more precise solution is 88.9972%. Vice versa the entropy corresponding to the exact probability 89% is approximately 0.499 916 Sh.

When we report a 50% and an 89% credibility interval we are therefore reporting intervals having uncertainties of one shannon and of half a shannon. Of course the arbitrariness remains.

For your amusement, some corresponding values of entropies and probabilities are reported in table 1.

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[7] ISO 2008 Items 13-24, 13-25. [8] https://electropedia.org/iev/iev.nsf/display?openform&ievref=171-07-11

entropy/Sh	probability/%
1.00	**50.00**
0.95	63.09
0.90	68.40
0.85	72.40
0.80	75.70
0.75	78.55
0.70	81.07
2/3	82.60
0.65	83.33
0.60	85.39
0.55	87.27

entropy/Sh	probability/%
0.50	**89.00**
0.45	90.59
0.40	92.06
0.35	93.42
1/3	93.85
0.30	94.68
0.25	95.83
0.20	96.89
0.15	97.85
0.10	98.70
0.05	99.44
0.00	100.00

probability/%	entropy/Sh
50	**1.0000**
51	1.0000
52	0.9990
53	0.9970
54	0.9950
55	0.9930
56	0.9900
57	0.9860
58	0.9810
59	0.9770
60	0.9710
61	0.9650
62	0.9580
63	0.9510
64	0.9430
65	0.9340
66	0.9250
67	0.9150
68	0.9040
69	0.8930
70	0.8810
71	0.8690
72	0.8550
73	0.8410
74	0.8270

probability/%	entropy/Sh
75	0.8110
76	0.7950
77	0.7780
78	0.7600
79	0.7410
80	0.7220
81	0.7010
82	0.6800
83	0.6580
84	0.6340
85	0.6100
86	0.5840
87	0.5570
88	0.5290
89	**0.5000**
90	0.4690
91	0.4360
92	0.4020
93	0.3660
94	0.3270
95	0.2860
96	0.2420
97	0.1940
98	0.1410
99	0.0808

Table 1 Some corresponding values of probabilities and entropies

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